Are Rindler quanta real?

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first quantization

Begin with a classical phase space S, a real vector space equipped with a sympletic form σ, and a one-parameter group $t \mapsto T_t$ of symplectomorphisms.

first quantization is equivalently:

- a linear embedding η of S into a Hilbert space $\mathcal{H}$ such that $\eta(T_t f) = U_t \eta(f)$.
- a complex structure J on S.
- a real inner product μ on S.

second quantization

second quantization takes the one-particle space $\mathcal{H}$ to Fock space

$$\mathcal{F}(\mathcal{H}) \;=\; \mathbb{C} \oplus \mathcal{H} \oplus (\mathcal{H} \otimes \mathcal{H}) \oplus \cdots \,.$$

Observables and symmetries on $\mathcal{H}$ also lift to observables and symmetries on $\mathcal{F}(\mathcal{H})$.

vacuum state $\Omega = 1 \oplus 0 \oplus 0 \oplus \cdots$

creation and annihilation operators

$$
\begin{aligned}
a_n^*(f)(f_1 \otimes \cdots \otimes f_{n-1}) &= f \otimes f_1 \otimes \cdots \otimes f_{n-1} \\
a_n(f)(f_1 \otimes \cdots \otimes f_{n-1}) &= \langle f, f_1 \rangle_J f_2 \otimes \cdots \otimes f_{n-1}
\end{aligned}
$$

$$
\begin{aligned}
a^*(f) &= a_1^*(f) \oplus \sqrt{2} a_2^*(f) \oplus \sqrt{3} a_3^*(f) \oplus \ldots \\
a(f) &= 0 \oplus a_1(f) \oplus \sqrt{2} a_2(f) \oplus \sqrt{3} a_3(f) \oplus \cdots
\end{aligned}
$$

number and field operators

$$
\begin{aligned}
N(f) &= a^*(f)a(f) \\
N &= \sum_{i=1}^{\infty} N(f_i) \\
\Phi(f) &= \frac{1}{\sqrt{2}}(a^*(f) + a(f)) \\
W(tf) &= \exp(it\Phi(f)) \\
\langle \Omega, W(f)\Omega \rangle &= e^{-\langle f, f \rangle_J / 4}
\end{aligned}
$$

the Weyl algebra

objective: capture the algebraic structure of the canonical commutation relations

$$Q_n P_m - P_m Q_n = -i\delta_{nm}\hbar I.$$

step 1 (exponentiate)

$$e^{iaQ}e^{ibP} = e^{-iab/2}e^{ibP}e^{iaQ}.$$

step 2 (collect) $W(a,b) = e^{iab/2}e^{ibP}e^{iaQ}$

$$W(x)W(y) = e^{-i\sigma(x,y)}W(x+y).$$

the Weyl algebra

fact: There is a unique smallest C^*-algebra $\mathcal{A}[S, \sigma]$ generated by unitary operators $\{W(x) : x \in S\}$ satisfying the Weyl commutation relations.

Regular representations of the algebra $\mathcal{A}[S, \sigma]$ correspond to representations of the CCR.

Fock representations of $\mathcal{A}[S, \sigma]$ correspond to:

- complex structures on (S, σ)
- real inner products on (S, σ)
- quasifree states on $\mathcal{A}[S, \sigma]$

inequivalent Fock representations

fact In the Minkowski vacuum representation π_M, the quasilocal algebra $\pi_M(\mathcal{A}[S,\sigma])''$ is type III.

corollary The Rindler and Minkowski vacuum representations are disjoint.

Fulling's theorem

Two Fock vacuum representations $(\mathcal{H}, \Omega)$ and $(\mathcal{H}', \Omega')$ of the CCRs are unitarily equivalent if and only if:

$$\langle \Omega' | N \Omega' \rangle < \infty$$

The Minkowski and Rindler vacuum representations are inequivalent. Therefore,

$$\langle \Omega_m | N_r \Omega_m \rangle = \infty$$

Chaiken's theorem

Let $(\mathcal{H}_m, \Omega_m)$ be the Minkowski vacuum representation of the CCRs. Then the Rindler number operator N_r cannot be defined on the Minkowski Hilbert space $\mathcal{H}_m$. In fact, there is no Hilbert space on which both N_m and N_r can be simultaneously defined.

corollary. The expression $N_m N_r - N_r N_m$ doesn't make any sense.

corollary. The expression $\langle \Omega_m | N_r \Omega_m \rangle = \infty$ also doesn't make sense!

- A quantity cannot take an infinite value.
- N_r cannot be applied to the vector Ω_m.